

nuveen

A TIAA Company

Design studio guide: Charts and tables

2019

Overview

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It is up to the designers discretion to ultimately generate creative in the current brand standard. This will be different depending on audience and layout, thus this overview is considered a guideline only.

Tables

Exhibit 3: Correlations between global real estate, stocks and bonds

(2007-2017)

	Global			
	Private real estate	Listed REITs	Stocks	Bonds
Private real estate				
Listed REITs	0.12			
Stocks	0.24	0.88		
Bonds	-0.28	0.03	-0.07	

Data as of 31 December 2017. Indexes represented: Global private real estate (MSCI Global Total Return), global listed REITs (FTSE EPRA/NAREIT), global stocks (FTSE All-World Index), global bonds (Bloomberg Barclays Global Aggregate Bond Index). It is not possible to invest in an index. Performance for indices does not reflect fees or transaction costs. Source: MacroBond

Line: .5pt | Cool gray 4

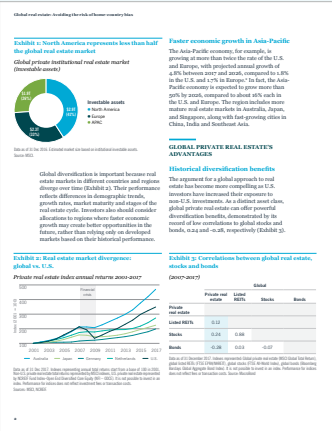
Font header: Newsgothic bold

Font body: Newsgothic Reg

Shading: 7% tint of blue brand colors for highlights, or Cool gray 4 at 10% tint.

NOTES:

- Limited space defaults to Newsgothic Condensed
- Colors are up to the discretion of the designer



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Nuveen Asset Management	AUM: \$188.4 billion
Fixed Income	
Municipal Fixed Income	
Short Term	● ▲ ■
Limited Maturity	● ▲ ■ +
Intermediate Term	● ▲ ■ +
Intermediate Term ESG	▲
Long Term	● ▲ ■ +
Strategic Muni Opps	● ■
High Yield	● ■ +
Short Duration High Yield	● ■
State Specific/State Preference	● ▲ ■ +
Taxable Municipals	● +
Inflation Protected	■

[illegible]

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	Nuveen Real Estate Securities Fund I FARCX	Nuveen Real Asset Income Fund I NRIIX	Nuveen Global Infrastructure Fund I FGIYX
Overall Morningstar Rating™	★★★★	★★★★★	★★★★
	Based on historical risk-adjusted total return among 225 Real Estate Funds	Based on historical risk-adjusted total return among 397 World Allocation Funds	Based on historical risk-adjusted total return among 75 Infrastructure Funds
Key fund characteristics			
Inception date	30 Jun 1995	13 Sep 2011	17 Dec 2007
Average annualized total returns (%)			
1-year	18.41	7.16	11.23
3-year	5.85	7.07	8.10
5-year	8.84	6.32	6.64
10-year	18.06		12.71

[illegible]

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Table - examples

Exhibit 8: TIPS Correlation to inflation is higher than many asset classes

Asset class	TIPS 1-10 year	TIPS							
TIPS: 1-10 year	1.00	TIPS							
TIPS: full duration	0.96	1.00	U.S. Invest grade bonds						
U.S. Invest. grade bonds	0.72	0.78	1.00	Foreign govt bonds					
Non U.S. Govt. Bonds	0.56	0.60	0.66	1.00	U.S. High yield bonds				
U.S. High-yield bonds	0.33	0.28	0.18	0.15	1.00	U.S. Stocks			
U.S. Equities	0.03	0.01	-0.09	0.05	0.62	1.00	Foreign stocks		
International equities	0.13	0.12	0.02	0.26	0.67	0.86	1.00	Real estate	
Real estate	0.26	0.25	0.20	0.23	0.60	0.59	0.57	1.00	Inflation
Inflation	0.17	0.08	-0.15	-0.05	0.12	0.03	0.05	0.05	

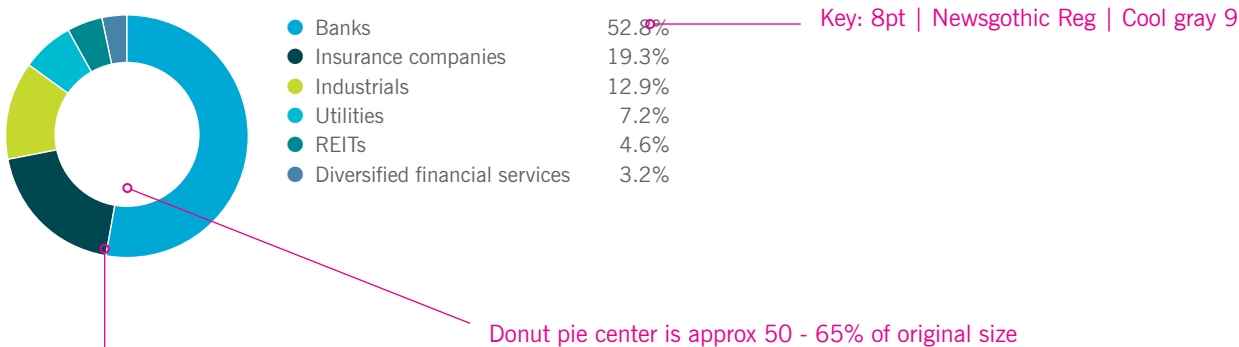
Data source: Morningstar Direct, Bloomberg, L.P. and Nuveen Asset Management, 01 Feb 1999 - 30 Jun 2018, longest time period available for the Bloomberg Barclays U.S. TIPS 1-10 Year Index. **Past performance does not guarantee future results. Representative Indexes:** **TIPS 1-10 Year:** Bloomberg Barclays U.S. Treasury TIPS 1-10 Year Index; **TIPS Duration:** Bloomberg Barclays U.S. Treasury TIPS Index; **U.S. Investment Grade Bonds:** Bloomberg Barclays U.S. Aggregate Bond Index; **Non U.S. Government Bonds:** Citigroup World Government Bond Index; **U.S. High Yield Bonds:** Bloomberg Barclays U.S. High Yield 2% Issuer Cap Index; **U.S. Equities:** S&P 500 Index; **International Equities:** MSCI EAFE Index; **Real Estate:** FTSE NAREIT All Equity REIT Index; **Inflation:** Consumer Price Index for All Urban Consumers, not seasonally adjusted.



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Pie charts

Exhibit 2: Financial institutions dominate the market



- NOTES:
- Limited space defaults to Newsgothic Condensed
 - Colors are up to the discretion of the designer
 - Legend location depends on layout: key is positioned top right of pie, or listed below

Exhibit 2: Financial institutions dominate the market

WHAT IS A PREFERRED SECURITY?
Preferred securities don't usually enter an asset allocation category on the capital markets of bank stocks and bonds. A preferred security can be classified as either debt or equity on the balance sheet, depending on its features. The easiest way to identify preferred securities is by their placement within the corporate capital structure.

Exhibit 2 illustrates how preferreds typically rank on the hierarchy between debt and equity. In a bankruptcy or liquidation, preferred security holders have a higher priority than common stock holders, but a lower priority than senior debt holders. They will be paid only if there is money left after senior creditors have been made whole.

Exhibit 2: Financial institutions dominate the market

Sector	Percentage
Banks	52.8%
Insurance companies	19.3%
Industrials	12.9%
Utilities	7.2%
REITs	4.6%
Diversified financial services	3.2%

TYPES OF PREFERRED SECURITIES
A preferred security's combination of features will classify it as either an equity or a debt security, but most preferred securities have elements of each. For example, some preferred securities generate income in the form of dividends, while others generate income in the form of dividends. Other common features are shown in Exhibit 3.

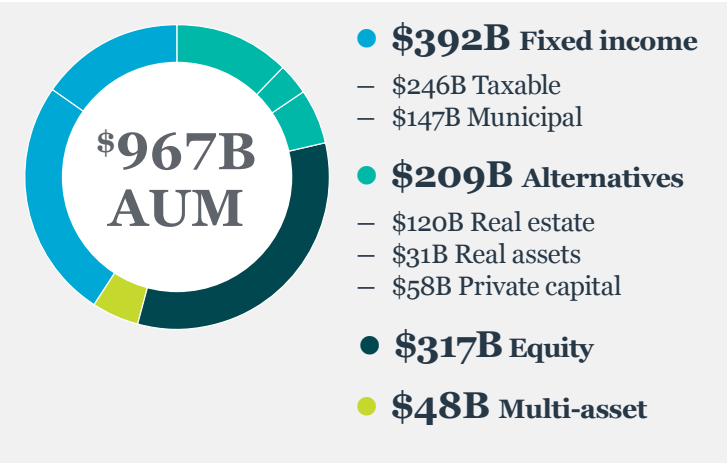
POTENTIAL BENEFITS OF PREFERRED SECURITIES
Preferred securities potentially offer relatively attractive yields, especially in today's lower rate environment. They may also provide low sensitivity to interest rate changes, portfolio diversification and tax advantages. This combination has created significant interest in the asset class.

Exhibit 2: Financial institutions dominate the market

A preferred security can be classified as either debt or equity on the balance sheet.

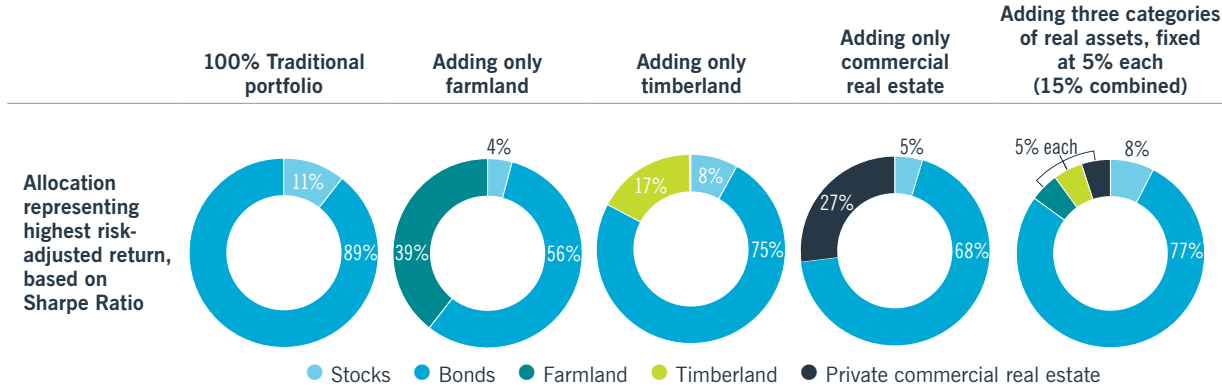
Write number: 2819_0818

Pie chart - examples



Wrike number: [6971_0419](#)

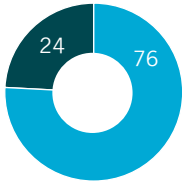
Comparing returns, standard deviations and Sharpe Ratios (1992 – 2017)



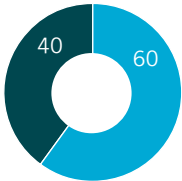
Wrike number: [3091_0818](#)

Pie chart - examples

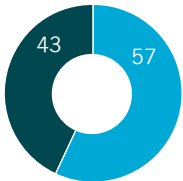
● Advanced economies ● Emerging markets and developing economies



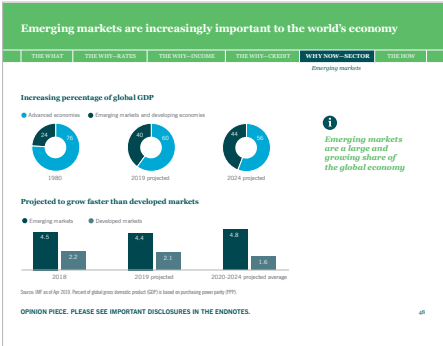
1980



2019 projected



2023 projected

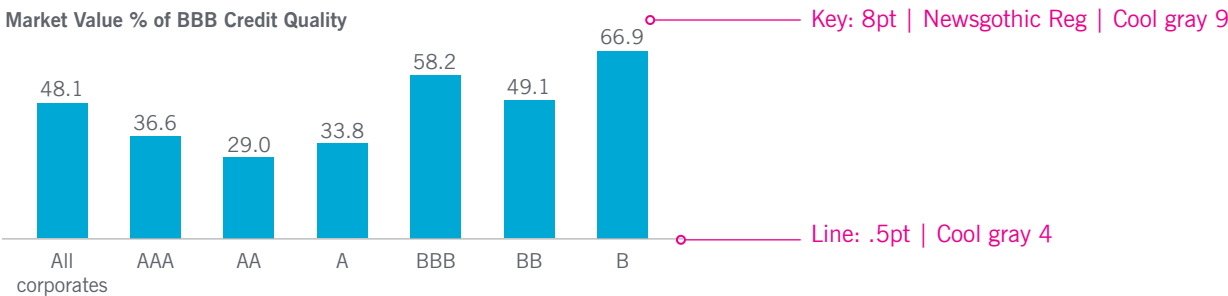


Wrike number: [6696_0419](#)

Bar charts

Exhibit: Yield and BBB credit quality composition by MSCI ESG rating (Issuer-weighted)

As of 30 Jun 2018



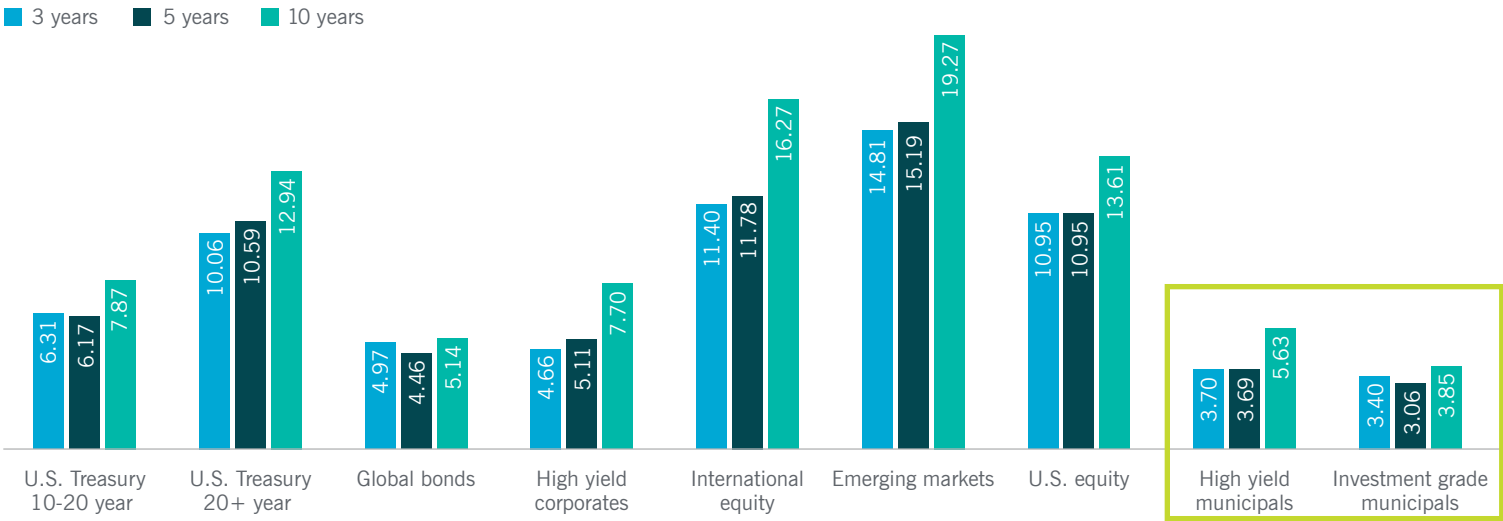
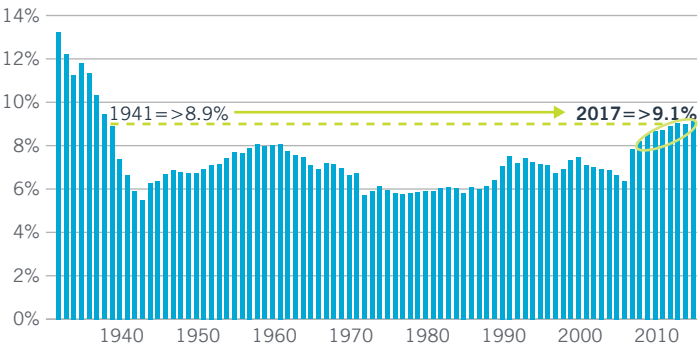
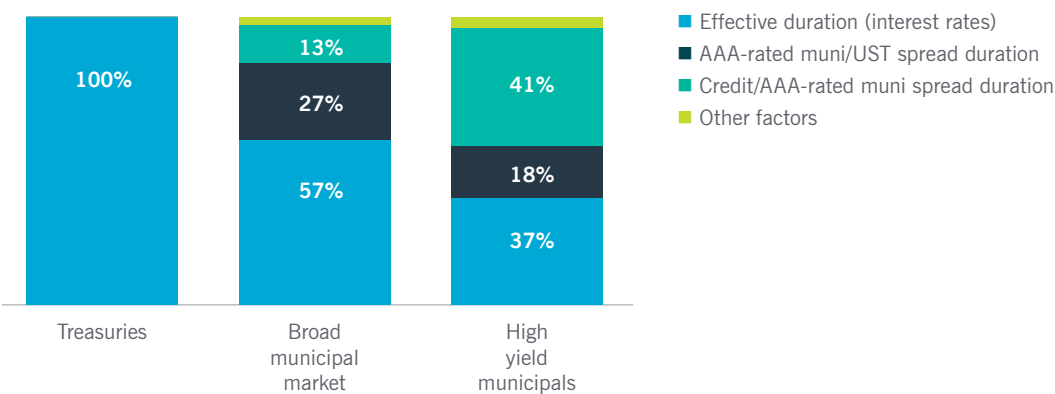
NOTES:

- Limited space defaults to Newsgothic Condensed
- Colors are up to the discretion of the designer
- Legend location depends on layout
- Labels for axes are not needed if placed in the header or subhead of chart
- Grid lines are not needed if data numbers are provided



Write number: 2819_0818

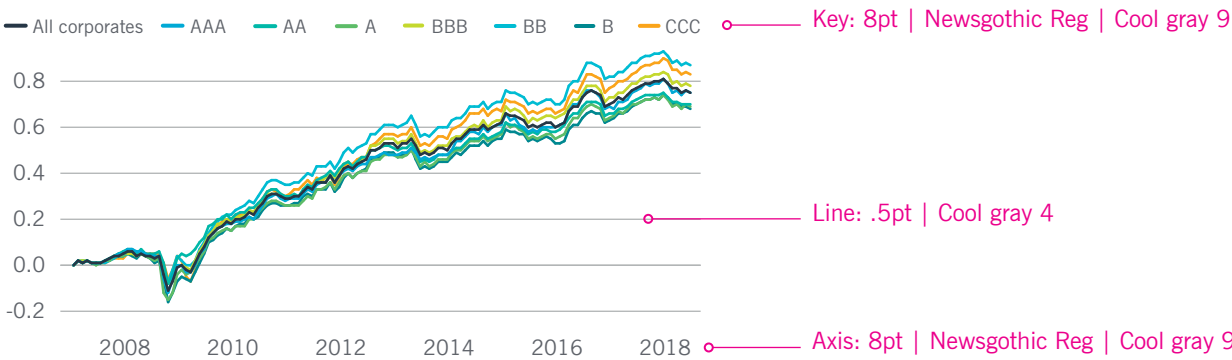
Bar chart - examples



Line charts

Exhibit 1: Cumulative total return by MSCI ESG rating (Issuance-weighted)

12/31/2006 - 6/30/2018



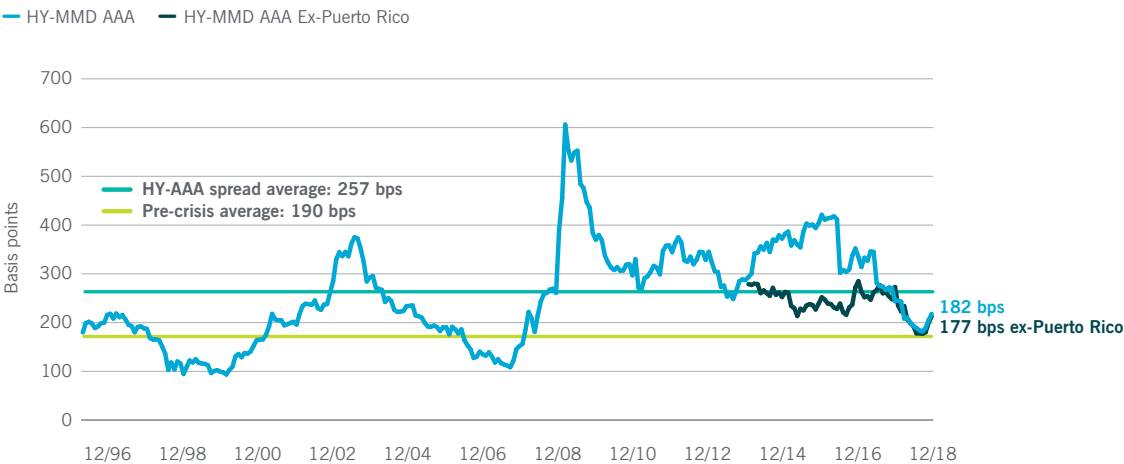
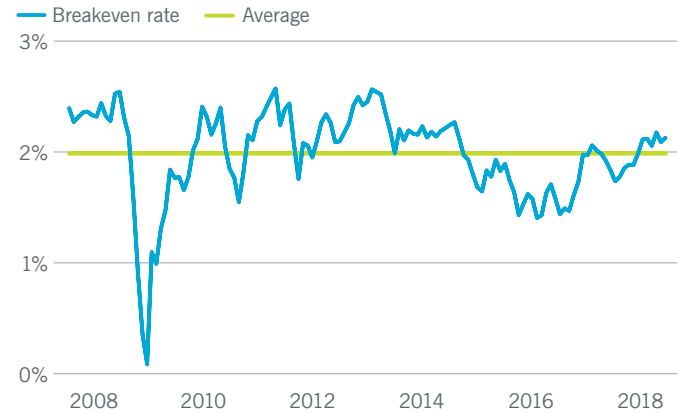
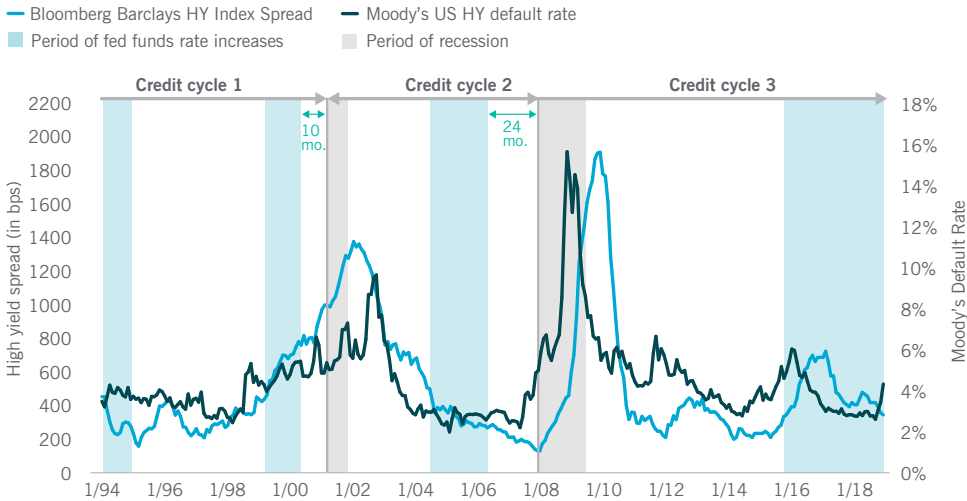
NOTES:

- Limited space defaults to Newsgothic Condensed
- Colors are up to the discretion of the designer
- Legend location depends on layout: with narrow columns the key is positioned upper left, wide columns the key is positioned top right
- Labels for axes are not needed if placed in the header or subhead of chart

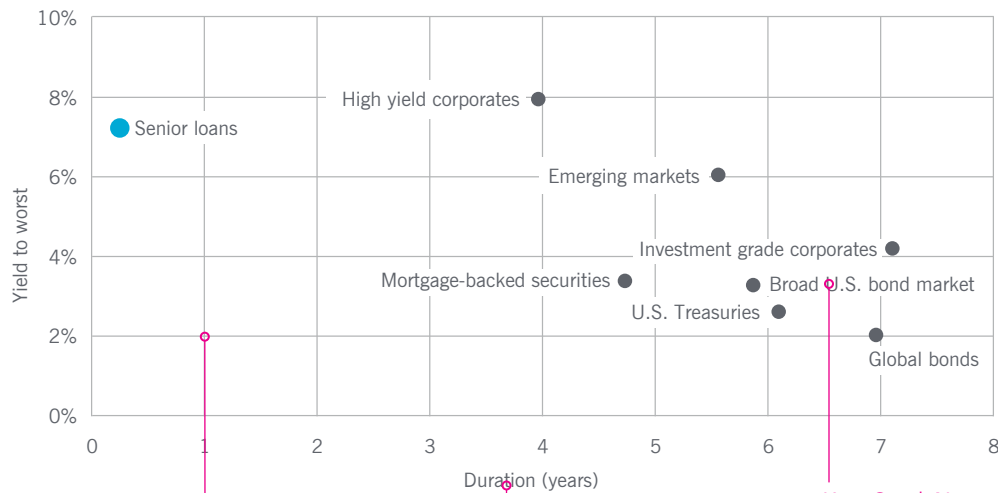


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Line chart - examples



Scatter and whisker charts



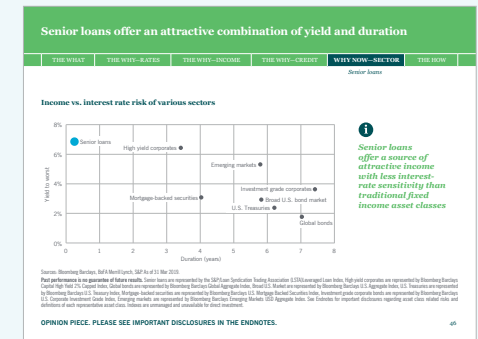
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Axis: 8pt | Newsgothic Reg | Cool gray 9

Gride Lines: .5pt | Cool gray 4

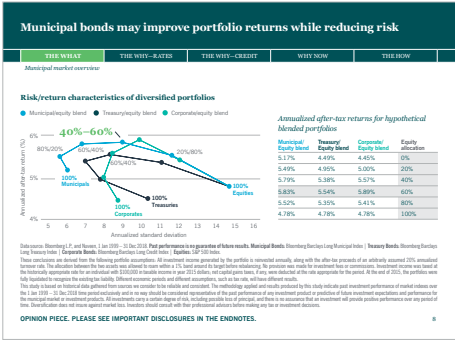
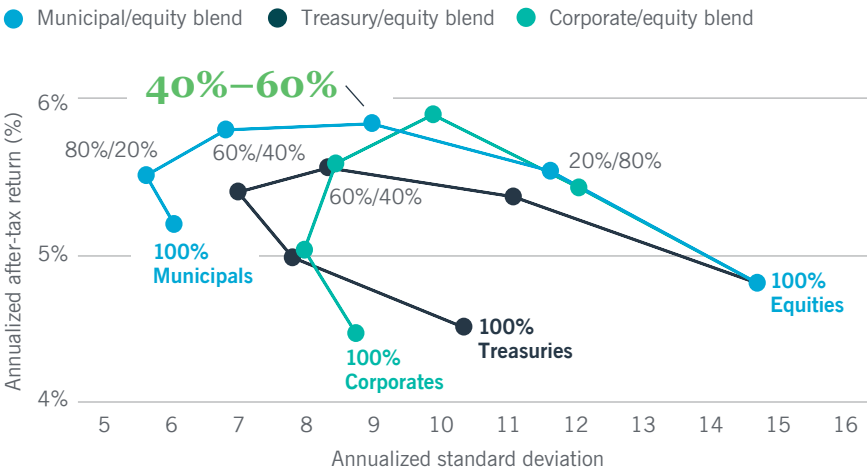
NOTES:

- Limited space defaults to Newsgothic Condensed
- Colors are up to the discretion of the designer - in some cases a selected color is used to highlight a particular point (as seen in this example)
- Legend location depends on layout
- Labels for axes are usually needed for scatter charts and centered relative to body of chart



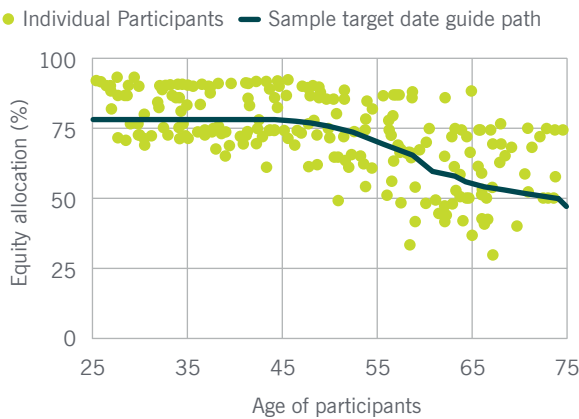
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Scatter and whisker chart - examples



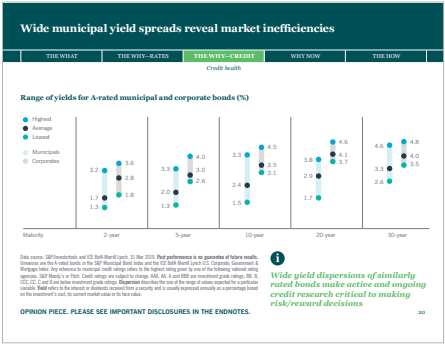
Write number: [6696_0419](#)

After re-enrollment Participants with more suitable allocations

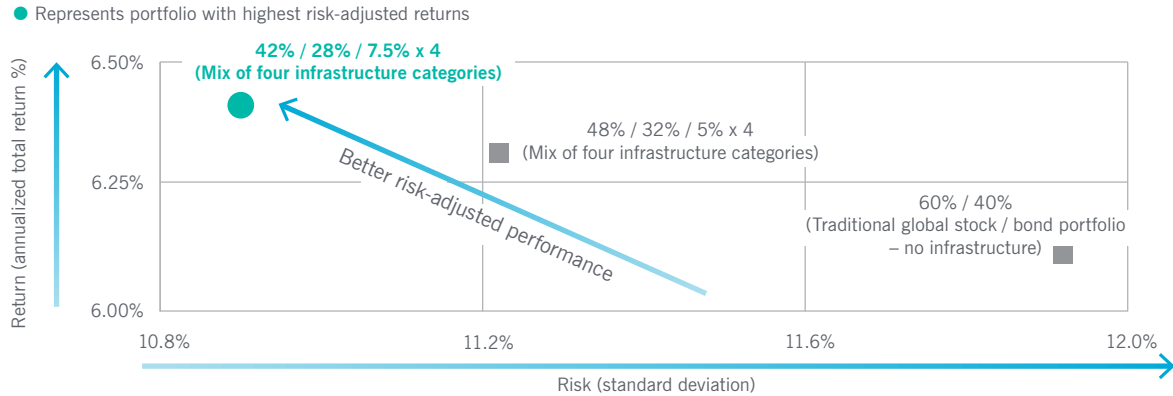


INTERNAL USE ONLY

Scatter and whisker chart - examples



Wrike number: [6696_0419](#)



INTERNAL USE ONLY

Color palette themes

